

Multi-Future Pedestrian Trajectory Forecasting with Intent–Trajectory Consistency Motivation

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Presentation goal: predict multiple plausible pedestrian futures rather than a single averaged future.

a) Problem: Pedestrian Future Motion is Multi-Modal

Task. Given a pedestrian's recent motion history in a driving video, predict their future motion.

Input:

$x_{1:15}$ = past 15 pedestrian center points

Output:

$\hat{y}_{1:45}$ = future 45 center points

In realistic scenes, the same observed history may lead to several plausible futures:

- continue walking,
- slow down,
- stop,
- start crossing.

Main difficulty: one-to-many prediction

A deterministic regression model tends to predict an “average” trajectory when several futures are possible.

One history $\longrightarrow$ multiple plausible futures

Project goal

Model pedestrian future uncertainty by predicting a set of K possible future trajectories.

b) Why This Problem Matters

Pedestrian forecasting is important for safe autonomous driving.

If the forecast is too late:

- vehicle may brake too late,
- unsafe near crosswalks,
- dangerous under occlusion.

If the forecast is too conservative:

- vehicle may stop unnecessarily,
- traffic flow becomes inefficient,
- behavior becomes uncomfortable.

Why multi-future prediction helps

Instead of committing to one future, the model can cover several possible futures and let downstream planning reason about uncertainty.

This is especially useful in ambiguous scenes, where a pedestrian may either keep walking or start crossing.

c) Approach: K-Hypothesis Trajectory Forecasting

We implemented a lightweight PyTorch prototype on PIE set03.

Data processing

- Parse PIE annotation XML files.
- Convert pedestrian bounding boxes to center points.
- Keep contiguous tracks with enough length.
- Construct sliding windows:

15 observed frames $\rightarrow$ 45 future frames.

Split

- Train: videos 0001–0014
- Validation: videos 0015–0017
- Test: videos 0018–0019

Model

- GRU encoder over past center trajectory.
- MLP prediction head.
- Output:

$$\hat{Y} = \{\hat{y}^{(1)}, \dots, \hat{y}^{(K)}\}, \quad \hat{y}^{(k)} \in \mathbb{R}^{45 \times 2}.$$

Best-of-K training loss

$$\mathcal{L}_{\text{BoK}} = \min_{k \in \{1, \dots, K\}} \frac{1}{T} \sum_{t=1}^T \left\| \hat{y}_t^{(k)} - y_t \right\|_2^2.$$

This encourages at least one predicted future to match the realized trajectory.

c) Evaluation Metrics and Baseline Comparison

Our main comparison is between the deterministic baseline and the multi-future model.

Baseline: $K = 1$

When $K = 1$, the model outputs only one future trajectory:

$$\hat{Y} = \{\hat{y}^{(1)}\}.$$

This degenerates to standard single-trajectory prediction. In practice, under one-to-many uncertainty, this tends to behave like predicting an averaged future.

Multi-future prediction: $K > 1$

When $K > 1$, the model outputs a set of possible futures:

$$\hat{Y} = \{\hat{y}^{(1)}, \dots, \hat{y}^{(K)}\}.$$

The goal is not necessarily to make the first trajectory best, but to ensure that at least one predicted future is close to the realized future.

d) Results: Multi-Future Improves Set Coverage

Main quantitative comparison on PIE set03 prototype.

Model	K	Epoch	ADE@1	FDE@1	minADE@K	minFDE@K
Baseline	1	30	13.27	34.68	13.27	34.68
Multi-future	5	30	15.11	39.48	7.17	16.30
Multi-future	10	20	60.32	202.90	6.40	13.05

Observation 1:

$$K = 5$$

substantially improves set-based metrics:

minADE : 13.27 $\rightarrow$ 7.17, minFDE : 34.68 $\rightarrow$ 16.30.

Observation 2:

ADE@1 and FDE@1 do not necessarily improve.

This is expected because Best-of-K optimizes whether *some* predicted future matches the ground truth, not whether the first/default future is best.

Key takeaway

Multi-future prediction improves **coverage** of possible futures, rather than simply improving a single deterministic trajectory.

d) Qualitative Results: Predicted Future Sets

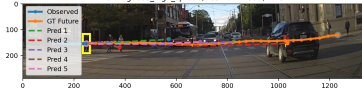
Success case

success_best_minADE, test idx=1314, K=5



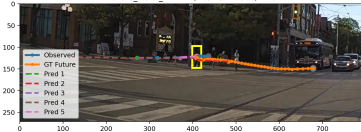
Ambiguous case

ambiguous_high_spread, test idx=1599, K=5



Failure case

failure_worst_minADE, test idx=1590, K=5



Interpretation. The predicted set can cover multiple plausible futures, but without mixture weights or an intention head, the model does not yet tell us which future should be trusted most.

e) Discussion: What We Learned

What worked

- A simple GRU-based model can generate multiple plausible future trajectories.
- Best-of-K training is effective for improving minADE/minFDE.
- Multi-future prediction is better suited than deterministic regression for one-to-many pedestrian motion.

What did not fully work

- The first/default trajectory may become worse even when minADE improves.
- The model currently has no learned probability ranking over futures.
- The final prototype does not yet include image context, road geometry, or vehicle speed.
- The full intent–trajectory consistency regularizer remains future work.

Main lesson

For ambiguous pedestrian behavior, evaluating only one predicted trajectory can be misleading. Set-based metrics reveal whether the model covers the realized future.

e) Future Directions

Next steps

1 Add mixture weights

$$\{\hat{y}^{(k)}\}_{k=1}^K \rightarrow \{(\pi_k, \hat{y}^{(k)})\}_{k=1}^K.$$

2 Add an intention head

$$p_{\text{cross}} = \Pr(\text{pedestrian will cross}).$$

3 Train intent–trajectory consistency

$$\mathcal{L}_{\text{cons}} = \left| p_{\text{cross}} - \sum_{k \in C} \pi_k \right|.$$

4 Use richer inputs

- pedestrian appearance, scene context,
- road/crosswalk geometry, ego-vehicle speed.

Final conclusion

Our prototype shows that multi-future forecasting can substantially improve trajectory coverage. The next important step is to make these futures semantically consistent with pedestrian intention.

Single future prediction → Multi-future coverage → Intent-consistent forecasting

Main result: Multi-future prediction improves minADE/minFDE, showing better coverage of ambiguous pedestrian futures.

Main limitation: Without mixture weights and an intention head, the model still lacks semantic consistency.